

Modeling with a Parent Function: the Open Box - Practice

Algebra II Honors | Unit 1 | Lesson 9 Practice | TEKS A2.2.A, A2.7.I

Objective: I can generalize the open-box model to any L-by-W sheet, expand it, and reason about average rate of change and scaling.

Section 1 — Recall

1. For a sheet L by W (W the shorter side), write $V(x)$ and its domain in terms of L and W.

$V(x) =$ _____ Domain: _____

2. For a 22-by-14 sheet, identify L, W, and give the contextual domain using the general formula.

L = _____ W = _____ Domain: _____

Section 2 — Apply

3. Expand $V(x) = x(16 - 2x)(10 - 2x)$ to standard form. Show the multiplication.

$V(x) =$ _____

4. For the 16-by-10 sheet, compute $V(1)$ and $V(6)$, and identify which x-value is inside the valid domain (0, 5).

$V(1) =$ _____ $V(6) =$ _____ Valid: _____

Section 3 — Challenge

5. A cube-shaped sheet ($L = W = s = 12$) has volume function $V(x) = x(12 - 2x)^2$. Find the average rate of change of V from $x = 1$ to $x = 2$.

Average rate: _____

6. For a general L-by-W sheet with W the shorter side, explain algebraically why $V(x) = x(L - 2x)(W - 2x)$ always has $x = 0$ as one of its zeros, regardless of L and W.

Section 4 — Explain It

7. A classmate argues that since volume multiplies three lengths together, doubling BOTH sheet dimensions (L and W) should exactly double the maximum volume. Use the scaling relationship (volume scales by k^3 when linear dimensions scale by k) to explain what's wrong.

